BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2024

CHEMISTRY

Past Paper 2024 • Paper Code: PR–X (A-I-B5) (24)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** PR–X (A-I-B5) (24)

Note: Four circles are given against each question. Fill only the circle of the correct answer. Use a blue or black marker/pen. Filling more than one circle will be treated as a wrong answer.

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. In which layer of atmosphere, weather balloons can fly;

(A) Troposphere
(B) Stratosphere
(C) Ozonosphere
(D) Mesosphere

Q2. Rock weathering is a major source for the production of;

(A) Oxides of nitrogen
(B) Oxides of carbon
(C) Oxides of Sulphur
(D) Volatile organic compounds

Q3. Which element is used for water disinfection?

(A) Fluorine
(B) Bromine
(C) Iodine
(D) Chlorine

Q4. Which of the given is the best method for disposal of solid waste?

(A) Land filling
(B) Recycling
(C) Burning
(D) Sea dumping process

Q5. Which of the given is the source of CO2 in manufacture of urea (NH2CONH2)?

(A) Natural gas
(B) Coal
(C) Petroleum
(D) Coke

Q6. Liquid form of petroleum is called as;

(A) Asphalt
(B) Crude oil
(C) Diesel
(D) Petrol

Q7. Expression (Kc) for the given reaction is: $3NO2 + H2O \rightleftharpoons 2HNO3 + NO$

(A) $[HNO3]^2[NO] / [NO2]^3[H2O]$
(B) $[HNO3]^2 + [NO] / [NO2]^3 + [H2O]$
(C) $[NO2]^3[H2O] / [HNO3]^2[NO]$
(D) $[NO2]^3 + [H2O] / [HNO3]^2 + [NO]$

Q8. NH3 is a Lewis base because it:

(A) Accepts pair of electron
(B) Donates pair of electrons
(C) Donates a proton
(D) Produces OH^- ion

Q9. Which one of the given is an organic compound?

(A) Carbon monoxide
(B) Carbon dioxide
(C) Bicarbonates
(D) Phenol

Q10. Which statement is true about organic compounds?

(A) Ionic in nature
(B) Conductor of electricity
(C) Thermally instable
(D) Thermally stable

Q11. First three members of alkynes are:

(A) Solid
(B) Liquid
(C) Gas
(D) Plasma

Q12. Two sugar units are linked to each other by;

(A) Peptide bond
(B) Glycosidic bond
(C) Sulphur bond
(D) Hydrogen bond

SECTION-B

Time Allowed: 2 Hours 45 Minutes

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q2. Answer any EIGHT parts. Each part carries FOUR marks.

i) Compare the Macroscopic characteristics of Forward Reaction and Reverse Reaction with reference to their mode of reaction.

ii) Calculate the pH and pOH of 0.079 M of HCl solution?

iii) Classify the given as saturated and unsaturated hydro carbons. CH4, C5H12, C3H6 and C4H6

iv) Convert Octane and Nonane to their alkyl group. Also write the name of alkyl groups formed.

v) How pentene is prepared from pentyl alcohols. Write complete reaction.

vi) Write down the chemical equations for the addition of HBr and HI to C2H4.

vii) How protein polymer is formed from amino acids. Give example.

viii) Complete the table below; Vitamin | Source | Deficiency Symptom — i. Vitamin-D | ii. Ascorbic Acid

ix) How acid rain affects on plants and trees?

x) Mention values of the following physical properties of water. (a) Latent heat of fusion (b) Latent heat of vaporization (c) Heat capacity (d) Density

xi) What do you know about enrichment of ore? Why it is necessary in metallurgical operations?

SECTION-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 16

> Note: Note: Attempt any THREE questions. All questions carry equal marks.

Q3.

(a) How equilibrium expression is written for given reaction. Derive its unit also. I2(s) + 5F2 ⇌ 2IF5

(b) What are the dissociation products of Mg(OH)2? Also categorize it as Arrhenius acid or base.

Q4.

(a) Explain two types of closed chain or cyclic organic compounds with examples.

(b) Write two steps involved in dehydrogenation f adjacent di-halides to prepare alkynes with conditions.

Q5.

(a) How oils and fats are produced from fatty acids. Explain with the help of chemical reaction?

(b) How depletion of Ozone layer affect the marine life and amphibians?

Q6.

(a) Explain Typhoid fever and dysentery as water born disease.

(b) Which fraction is represented by Carbon atoms (C14 to C25). Also write at least two uses.